

CS 643, Cloud Computing

Programming Assignment 2

Links:

Github: <https://github.com/qaz216/wineTrainingApp/tree/main>

Dockerhub: <https://hub.docker.com/repository/docker/qaz216/wine-training-app/general>

Setting Up EMR Spark Cluster:

- Go to Amazon EMR for EC2 Clusters

[Amazon EMR](#) > **EMR on EC2: Clusters**

- Click on “Create Cluster”



- Name Cluster

Name

Spark Wine Training Cluster

- Use latest EMR release for Spark cluster

Amazon EMR release [Info](#)

A release contains a set of applications which can be installed on your cluster.

emr-7.5.0

Application bundle

Spark Interactive	Core Hadoop	Flink	HBase	Presto	Trino	Custom
						
☐ AmazonCloudWatchAgent 1.300032.2	☐ Flink 1.19.1	☐ HBase 2.5.10	☐ Hive 3.1.3	☐ JupyterHub 1.5.0	☐ Phoenix 5.2.0	☐ Spark 3.5.2
☐ HCatalog 3.1.3	☒ Hadoop 3.4.0	☐ JupyterEnterpriseGateway 2.6.0	☐ Tez 0.10.2	☐ ZooKeeper 3.9.2		
☐ Hue 4.11.0	☐ Oozie 5.2.1					
☒ Livy 0.8.0	☐ Presto 0.287					
☐ Pig 0.17.0						
☐ TensorFlow 2.16.1						
☐ Zeppelin 0.11.1						

- Choose a custom AMI for desired Ubuntu image

Operating system options [Info](#)

- ☐ Amazon Linux release
- ☒ Custom Amazon Machine Image (AMI)

Custom AMI

X

- Select “Uniform Instance Groups” for cluster configuration

▼ Cluster configuration - *required* [Info](#)

Choose a configuration method for the primary, core, and task node groups for your cluster.

- ☒ **Uniform instance groups**
Choose the same EC2 instance type and purchasing option (On-Demand or Spot) for all nodes in your node group. [Learn more](#)
- ☐ **Flexible instance fleets**
Choose from the widest variety of provisioning options for the EC2 instances in your cluster. Diversify instance types and purchasing options, and use an allocation strategy. [Learn more](#)

- Remove Task Instance Group

Task 1 of 1

Remove instance group

Name

Task - 1

Choose EC2 instance type

m5.xlarge
4 vCore 16 GiB memory
EBS only storage On-Demand price: -
Lowest Spot price: -

Actions ▼

- Set instance size to 4

Provisioning configuration

Set the size of your core instance group. Amazon EMR attempts to provision this capacity when you launch your cluster.

Name	Instance type	Instance(s) size	Use Spot purchasing option
Core	m5.xlarge	4	☐

- Provision VPC & Subnet

▼ Networking - *required* [Info](#)

Choose the network settings that determine how you and other entities communicate with your cluster.

Virtual private cloud (VPC) [Info](#)

vpc-02d6dc3dd5cc47a88

Browse

Create VPC

Subnet [Info](#)

subnet-0017cce3e45f192db

Browse

Create subnet

- Create security configuration and key pair for cluster

▼ **Security configuration and EC2 key pair** [Info](#)

Choose a security configuration or create a new one that you can reuse with other clusters.

Security configuration

Select your cluster encryption, authentication, and instance metadata service settings.

Amazon EC2 key pair for SSH to the cluster [Info](#)

- Select a service role

Amazon EMR service role [Info](#)

The service role is an IAM role that Amazon EMR assumes to provision resources and perform service-level actions with other AWS services.

☒ **Choose an existing service role**
Select a default service role or a custom role with IAM policies attached so that your cluster can interact with other AWS services.

☐ **Create a service role**
Let Amazon EMR create a new service role so that you can grant and restrict access to resources in other AWS services.

Service role

- Create cluster

Create cluster

- Verify that “instance groups” have been created

☐	☐ Primary	ig-2QVJ7YKBL6CKO	☒ Running	1
☐	☐ Core	ig-1JRXEANZGNSXG	☒ Running	4

- Upload “TrainingDataset.csv” & “ValidationDataset.csv” to S3

☐	☐ TrainingDataset.csv	csv	December 5, 2024, 16:33:53 (UTC-05:00)	65.9 KB	Standard
☐	☐ ValidationDataset.csv	csv	December 5, 2024, 16:33:53 (UTC-05:00)	8.5 KB	Standard

- Log into primary server and clone code

```
ubuntu@ip-172-31-86-25:~$ git clone https://github.com/qaz216/wineTrainingApp.git
```

- ‘cd’ into wineTrainingApp and open configuration file

```
vi src/main/resources/application.properties
```

- Set application mode to 'train_model'

```
# application mode  
app.mode=train_model
```

- Set Master Node to "yarn" in config file

```
# (While running on EMR)  
app.master.node=yarn  
#app.master.node=local\[*\]
```

- Set S3 bucket location

```
# S3 bucket  
app.s3.bucket=s3a://project2-s3-bucket/
```

- Set AWS "ACCESS_KEY", "SECRET_KEY" & "SESSION_TOKEN" in config file

```
# NJIT AWS Credetials  
app.aws.access.key=<AWS ACCESS KEY>  
app.aws.secret.key=<AWS SECRET KEY>  
app.aws.session.key=<AWS SESSION KEY>
```

- Package & create JAR file

```
~/wineTrainingApp$ mvn clean package
```

- Use jar file: target/wine-training-app-1.0-SNAPSHOT.jar

```
./target/wine-training-app-1.0-SNAPSHOT.jar
```

- Upload JAR to S3

S3 URI
 s3://project2-s3-bucket-aryeh/wine-training-app-1.0-SNAPSHOT.jar

- Goto Cluster Instance & "Add Step"

Steps (0) [🔗](#)

❌ Failed	0
🔄 Running	0
✅ Completed	0
⏸ Canceled	0
⏳ Pending	0
⏴ Cancel pending	0
⏶ Interrupted	0

[Add step](#)

- Select “Spark Application” for setting type

☒ **Spark application**
 Adds a step that submits work to the Spark framework on the cluster.

- Add Jar Location

JAR location
 The JAR location may be a path into S3 or a fully qualified java class in the classpath.

- Add the following to “Spark-submit options”

Spark-submit options - optional
 Specify other options for spark-submit.

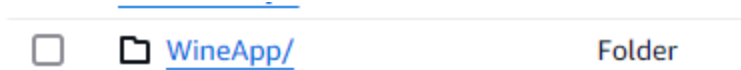
- Click “Add Step” button

[Add step](#)

- Verify that step results in “Completed” status

☐ ☒ **s-04209343L3H92K66VEIO** ✅ Completed

- Verify that results folder has been created in bucket



Running Model (Maven):

- Create EC2 Instance



- Name instance

Name and tags [Info](#)

Name

Wine Training Application

- Select Ubuntu 24 AMI

Amazon Machine Image (AMI)

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type
ami-0e2c8caa4b6378d8c (64-bit (x86)) / ami-0932ffb346ea84d48 (64-bit (Arm))
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

- Launch Instance

Launch instance

To get started, launch an Amazon EC2 instance, which is a virtual server in the cloud.



- Install Maven on EC2 instance

```
sudo apt-get install software-properties-common
sudo apt-add-repository universe
sudo apt-get update
sudo apt-get install maven -y
```

- Install docker in instance

Instructions here: <https://docs.docker.com/engine/install/ubuntu/>

- Clone git repository

```
git clone https://github.com/qaz216/wineTrainingApp.git
```

- Cd to wineTrainingApp directory

```
cd wineTrainingApp/
```

- Compile code

```
mvn clean compile
```

- Run application

```
mvn exec:java -Dexec.mainClass="com.njit.wineapp.WineTrainingApp"
```

- View results output to standard output

```
weightedPrecision: 0.5629104116222761  
weightedRecall: 0.575  
accuracy: 0.575  
f1: 0.5672726692311375
```

Running Model (Docker):

- Compile code, build docker image and push to docker hub (local computer)

```
docker build -t qaz216/wine-training-app .
```

- Run docker (from instance)

```
sudo docker run -it --name wine-training-app qaz216/wine-training-app
```

- Verify F1 through standard output

```
weightedPrecision: 0.5629104116222761  
weightedRecall: 0.575  
accuracy: 0.575  
f1: 0.5672726692311375
```

